30 Neighbourhood Watt: Collaborative Optimisation of Solar and Storage

External Resource Person

Key Topics / Themes

Distributed Energy Resources, Optimisation, Grid Services, Multi-Agent Systems

Problem Space/Domain

Australia's energy transition is increasingly driven by the widespread adoption of rooftop solar and residential batteries. While these distributed energy resources empower households to generate and store their own electricity, uncoordinated operation can cause issues such as local voltage fluctuations, reverse power flows, and reduced efficiency of network infrastructure.

At the same time, this growing network of prosumers presents a powerful opportunity: if homes and communities coordinate their energy behaviours, rather than acting independently, they can collectively reduce stress on the grid, improve self-sufficiency, and even access new value streams such as peer-to-peer trading or grid support services.

This project explores collaborative control strategies, where solar and storage assets dynamically coordinate their behaviour via APIs and optimisation models. Instead of static or individual decisions, devices operate as part of a larger system, sharing information, responding to common constraints, and adjusting in real time to achieve shared goals. This is an emerging area at the intersection of smart grid technologies, multi-agent systems, and sustainable energy management.

Project Brief

This project invites students to design and prototype a system that coordinates rooftop solar and battery storage across a neighbourhood or small community. Rather than each household acting independently, the goal is to explore how distributed energy assets can work together to improve local outcomes, such as reducing peak demand, increasing energy self-consumption, or smoothing grid impacts.

Students will simulate a network of homes with solar and/or storage, each with varying loads, tariffs, weather conditions, and user preferences. Using a combination of APIs, optimisation models, and coordination strategies, the system should allow these assets to communicate and adapt their behaviour in real time. Coordination approaches might include rule-based systems, agent-based modelling, cooperative optimisation, or lightweight predictive controls.

The final prototype should provide both technical functionality and visual feedback, showing how collaboration improves energy outcomes compared to uncoordinated behaviour. It should also demonstrate flexibility in handling different user goals, such as cost saving, carbon reduction, or backup priorities, and reflect real-world constraints such as network limits or time-of-use pricing. This project sits at the intersection of energy systems, distributed control, and digital coordination, offering students the opportunity to explore intelligent infrastructure at the community scale.

Students should consider technologies such as Python with Pyomo or Gurobi for optimisation, open-source distributed energy resource management simulators such as OpenDSS or GridLAB-D, EnergyPlus or other public datasets for realistic load and solar profiles, and web-based visualisation tools such as Plotly, Dash, or React. Useful skills to develop include optimisation and algorithm design, familiarity with energy systems or Python simulation frameworks, and front-end or UX skills for building visual dashboards. Relevant background material includes national integrated system planning publications, distributed energy resource research, open power systems datasets, and Tushar, Saha, Yuen, Smith, and Poor's overview of peer-to-peer trading in electricity networks.

Success Criteria

A successful project should demonstrate:

- A simulated network of solar and battery prosumers with realistic behaviour, modelling multiple households with rooftop solar and/or batteries, incorporating key variables such as load demand, PV generation, tariffs, and storage behaviour over time.
- An implemented coordination strategy, such as rule-based, agent-based, or predictive control, that demonstrates improved energy outcomes through collaboration compared to uncoordinated baselines.
- Clear, time-based visualisations showing how coordination reduces peak export or import at the grid

connection point, illustrating system-wide benefits such as improved self-consumption or smoother demand profiles.

- Support for diverse user goals in the optimisation process, allowing variation in household priorities such as lowest cost, highest autonomy, or lowest emissions, and demonstrating how these preferences influence collective optimisation outcomes.